

LIAWENEE, Australia

WMO: 959590

Lat: 41.8997S

Lon: 146.6694E

Elev: 1056

StdP: 89.27

Time Zone: 10.00 (AUS)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-5.7	-4.3	-8.3	2.1	4.0	-6.1	2.6	3.0	15.1	0.7	13.3	1.1	1.1	270	0.768

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.7	24.4	13.8	22.3	13.1	20.5	12.5	15.7	21.0	14.7	19.5	13.7	18.4	6.5	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	11.3	16.6	12.9	10.5	15.9	11.9	9.8	15.1	47.5	21.2	44.6	19.4	41.8	18.5	19.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.4	10.3	DB	-8.6	27.8	1.0	2.1	-9.3	29.4	-10.0	30.6	-10.5	31.8	-11.3	33.4	(4)
(5)			WB	-9.3	18.0	1.2	1.0	-10.2	18.7	-10.9	19.3	-11.5	19.9	-12.4	20.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	11.6	11.2	9.5	6.7	4.4	2.6	1.8	2.1	3.7	5.8	8.3	9.7	(6)	
	DBStd	4.57	3.99	3.30	3.61	2.94	2.63	2.26	2.07	2.38	2.48	3.30	3.55	3.42	(7)	
	HDD10.0	1547	27	23	52	106	174	223	254	243	189	137	73	47	(8)	
	HDD18.3	4349	214	201	273	349	432	473	512	502	438	387	299	267	(9)	
	CDD10.0	247	76	56	38	7	1	0	0	0	1	7	24	38	(10)	
	CDD18.3	6	4	1	1	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	109	69	11	13	0	0	0	0	0	0	0	3	13	(12)	
	CDH26.7	14	12	2	1	0	0	0	0	0	0	0	0	1	(13)	
(14) Wind	WSAvg	5.1	5.3	4.7	4.6	4.7	5.2	4.6	5.0	5.3	5.6	5.5	5.4	5.4	(14)	
(15) Precipitation	PrecAvg	1209	75	58	72	84	107	122	151	159	131	101	84	80	(15)	
	PrecMax	1561	183	150	165	175	255	258	219	296	228	189	159	167	(16)	
	PrecMin	807	13	10	28	32	32	23	84	46	51	36	15	25	(17)	
	PrecStd	146	45	30	37	36	60	49	41	63	48	36	41	35	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.3	24.7	25.0	18.6	14.6	10.9	8.5	12.0	14.6	20.5	23.2	25.6	(19)	
		MCWB	16.0	14.4	13.9	11.8	8.0	6.5	6.0	5.7	7.9	10.7	12.3	14.3	(20)	
	2%	DB	25.3	22.4	21.6	16.0	11.7	8.8	7.0	9.3	12.2	17.6	20.7	22.0	(21)	
		MCWB	14.3	13.5	13.1	10.2	7.7	6.6	5.0	5.8	6.9	8.9	11.7	12.4	(22)	
	5%	DB	22.6	20.5	19.1	14.1	10.1	7.5	6.1	7.6	10.1	14.9	18.5	19.6	(23)	
		MCWB	13.4	13.2	12.4	9.4	7.5	5.9	4.5	5.1	6.7	8.6	10.9	11.8	(24)	
	10%	DB	20.4	18.5	16.9	12.2	8.9	6.5	5.2	6.4	8.6	12.7	16.2	17.4	(25)	
		MCWB	12.6	12.4	11.5	9.0	7.1	5.2	4.0	4.7	5.9	7.9	10.4	11.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.8	16.5	16.0	12.7	10.3	8.8	7.1	7.6	9.9	12.7	14.6	16.0	(27)	
		MCDB	25.0	21.1	20.9	15.9	12.1	9.5	7.7	9.4	11.8	16.1	19.7	20.8	(28)	
	2%	WB	15.9	15.2	14.6	11.5	9.0	7.3	5.8	6.6	8.4	11.2	13.2	14.3	(29)	
		MCDB	22.2	19.8	18.8	14.0	10.3	8.1	6.7	8.5	10.7	14.9	18.0	19.0	(30)	
	5%	WB	14.6	14.3	13.5	10.5	8.2	6.4	5.0	5.8	7.3	9.8	12.1	13.0	(31)	
		MCDB	20.4	18.5	17.5	13.1	9.6	7.4	5.9	7.4	9.7	13.5	16.8	17.4	(32)	
	10%	WB	13.4	13.3	12.4	9.5	7.4	5.5	4.4	5.0	6.5	8.6	11.0	11.9	(33)	
		MCDB	18.8	17.2	16.2	12.0	8.8	6.6	5.3	6.5	8.7	12.3	15.2	16.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	11.7	11.3	9.1	7.4	6.6	6.0	6.8	8.4	10.4	11.2	11.5	(35)	
		MCDBR	16.4	15.1	15.1	13.0	9.7	9.0	7.5	9.6	11.4	14.9	15.3	15.6	(36)	
	5% WB	MCWBR	8.3	8.1	8.4	7.5	6.2	6.0	5.6	6.2	7.2	8.3	8.4	8.7	(37)	
		MCWBR	14.0	12.2	12.4	9.7	7.3	6.8	6.1	7.2	9.4	11.7	12.9	12.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.302	0.295	0.287	0.278	0.266	0.257	0.259	0.272	0.289	0.293	0.299	0.296	(40)	
		taud	2.589	2.625	2.645	2.641	2.652	2.663	2.664	2.630	2.566	2.569	2.572	2.600	(41)	
	Ebn at Noon		1028	1009	976	917	864	841	865	912	956	999	1023	1039	(42)	
		Edh at Noon	102	93	83	71	59	53	57	70	88	97	103	102	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.79	5.78	4.35	2.94	1.85	1.44	1.60	2.38	3.52	5.02	6.07	6.64	(44)	
	RadStd		0.49	0.28	0.21	0.18	0.13	0.13	0.09	0.19	0.24	0.29	0.47	0.45	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	+0.32	N/A	N/A	+0.67	+0.33	N/A	N/A	-98	+106	+12

Nomenclature: See separate page